Tiny Recursive Models vs. Fine-Tuned LLMs for Structured Reasoning: Sudoku and Maze Benchmarks

Ahmed AlShamy , **Armin Ghaseminejad** , **Nickolas Greiner**
School of Computing and Creative Technologies, University of the West of England, Bristol
Module: Machine Learning (UFCFAS-15-2)
Module Leader: Professor Jun Hong

Abstract

Solving puzzles like Sudoku requires working through steps one at a time, checking and correcting choices along the way. This is difficult for standard neural networks because they mostly rely on spotting patterns rather than careful step-by-step reasoning. Despite their scale, popular LLMs fail entirely on such tasks: GPT-2 Small (124M), Qwen2.5-0.5B (500M), and Llama-3.2-1B (1B) all achieve 0% puzzle-level accuracy on Sudoku-Extreme. This project investigates whether a Tiny Recursive Model (TRM), a recently proposed architecture designed for recursive problem-solving, can effectively solve these puzzles. The TRM works by repeatedly improving its answer step by step using learned recursive updates, with only about 5–7 million parameters. We train and evaluate the model on Sudoku-Extreme and Maze-Hard benchmarks, comparing performance against fine-tuned and distilled LLM baselines. Model performance is evaluated using puzzle accuracy, cell accuracy, and environmental metrics, including energy consumption and CO_2 emissions. We replicate key ablation results from Jolicoeur-Martineau [2025] on recursion depth and iteration limits. Results show that recursive architectural design, rather than parameter count, is the critical factor for constraint-satisfaction tasks, and that distilled student models offer a surprisingly competitive efficiency-accuracy trade-off.

1 Introduction

Complex puzzle problems like Sudoku and maze solving have always been difficult for machine learning systems. Unlike classification or text generation tasks, these problems require a model to follow rules, recover from mistakes, and plan several steps ahead, which makes them a poor fit for pattern recognition alone [Wang et al., 2024]. Large language models have made impressive strides on language and coding benchmarks, but raw scale does not seem to be enough. Fine-tuned LLMs spanning 124M to 1B parameters — GPT-2 Small, Qwen2.5-0.5B, and Llama-3.2-1B — all score 0% on Sudoku-Extreme, a benchmark built around minimal-clue puzzles that force deep recursive reasoning [Jolicoeur-Martineau, 2025].

Common ways of trying to fix this problem include using larger models, increasing context windows, and chain-of-thought prompting [Wei et al., 2022]. These methods can help on some benchmarks, but they are expensive and tend to break down wherever strict logical accuracy is required. That gap raises a more pointed question: can a small, purpose-built model actually beat frontier-scale systems on recursive reasoning, and do so at a fraction of the computational cost?

Jolicoeur-Martineau [2025] recently introduced the Tiny Recursive Model (TRM) as a possible solution. Instead of making the model bigger, TRM improves its answer step by step, refining internal states and possible solutions over several passes, doing multi-step reasoning with only 5–7 million parameters. Early results on Sudoku-Extreme and Maze-Hard were impressive, with TRM performing better than much larger models. Beyond benchmarks, this has direct real-world relevance: on an embedded device or edge server a 500M-parameter LLM simply cannot run, whereas a 5M TRM solving the same constraint-satisfaction problem fits comfortably within edge energy and memory budgets (Section 5.2).

This project reproduces and extends those findings. We evaluate TRM-MLP on Sudoku-Extreme and TRM-Att on Maze-Hard against four fine-tuned LLM baselines (GPT-2 Small, SmolLM2-360M, Qwen2.5-0.5B, Llama-3.2-1B) and two distilled student models, measuring accuracy alongside energy and CO_2 via CodeCarbon [Courty et al., 2023]. The published TRM checkpoint reaches 84.74% on Sudoku and 79.60% on Maze-Hard; our fine-tuned replication reaches 74.54%; every LLM baseline scores 0% puzzle-level, confirming that recursive structure, not parameter count, is the critical factor.

Original contributions: **(1) Energy survey** — first systematic energy-and-accuracy comparison across TRM, fine-tuned LLMs, and distilled students under identical hardware; **(2) Distillation finding** — a 2.4M student trained in 3 min achieves 35.87% cell accuracy, nearly 2× its 500M teacher; **(3) Artefact fix** — discovery and correction of a `mask_non_path` bug that inflated all Maze-Hard LLM baselines to 100%; **(4) K-vote study** — $K \in \{1, 2, 4\}$ majority voting confirms test-time compute does not substitute for recursive inductive bias.

2 Related Work

2.1 Hierarchical and Tiny Recursive Reasoning Models

Wang et al. [2024] introduce the Hierarchical Reasoning Model (HRM), which uses two small recurrent networks at different frequencies with deep supervision to reach 55% puzzle accuracy on Sudoku-Extreme with 27M parameters, substantially outperforming standard LLM baselines (0% puzzle accuracy). Jolicoeur-Martineau [2025] proposes TRM as a simpler, more parameter-efficient replacement: a single 2-layer network with effective depth 42 at only 5–7M parameters, reaching 87.4% on Sudoku-Extreme. Key ablations show EMA is critical, recursion depth matters, and MLP-Mixer outperforms self-attention on the fixed 81-token Sudoku context. These findings directly inform our training protocol (Section 4).

2.2 Test-Time Adaptation and WTA Training

McGovern [2025] shows that a pre-trained TRM can be fine-tuned in only 12,500 gradient steps on unseen tasks, orders of magnitude less than training from scratch; full fine-tuning substantially outperforms LoRA because TRM's low parameter count makes adapters unnecessarily restrictive. This contrast motivates our choice to use LoRA for LLM baselines while training TRM without adapter restrictions. Dillon [2026] extends TRM with Winner-Takes-All (WTA) training, running K parallel latent initialisations that compete during training, reaching 96.9% $\pm$ 0.6% on Sudoku-Extreme with $K = 4$ heads and showing ACT halting step correlates with puzzle difficulty ($r = 0.73$). Our K-vote experiment (Section 4.4) tests naive majority voting without WTA, and the null result confirms Dillon's [2026] finding that latent diversity must be internalised during training.

2.3 Tiny Autoregressive Recursive Models

Rauba et al. [2026] critically re-evaluate TRM in autoregressive settings, finding no reliable performance gains over simpler baselines under compute-matched comparisons. Their critique does not apply to our work (supervised non-autoregressive setting), but it delineates where TRM-style models are most effective: constrained, fixed-output problems with iterative refinement.

2.4 Knowledge Distillation

Knowledge distillation (KD) [Hinton et al., 2015] trains a small student on the soft probability distributions produced by a larger teacher rather than on one-hot labels. Soft targets encode inter-class similarity: for constraint-satisfaction tasks the combinatorial output structure means near-miss cell assignments carry residual probability mass, exposing structure that hard labels discard. Sanh et al. [2019] demonstrate the principle transfers to language model compression; we apply the same KD objective to our distilled LLMs (§4).

3 Data

3.1 Sudoku-Extreme

Sudoku-Extreme is a benchmark of constraint-satisfaction problems introduced by Jolicoeur-Martineau [2025], originally derived from the Palm et al. [2018] dataset. Each puzzle is a 9×9 Sudoku grid with exactly 17 clues (the minimum to guarantee a unique solution), making the problem extremely difficult. The dataset contains 1,000 training samples and 423,000 test samples.

Each puzzle is encoded as a flat integer sequence of length 81 for TRM (with a learnable $[0, 1, D]$ positional embedding) and as an 81-character digit string with task prefix for LLMs; both encodings are consistent across all model families so comparisons reflect architecture rather than representation. To compensate for the small training set, 1,000 digit permutation augmentations are generated per example, preserving Sudoku validity.

3.2 Maze-Hard

Maze-Hard is a pathfinding benchmark introduced by Jolicoeur-Martineau [2025], generated via the procedure of Lehnert et al. [2024]. Each maze is a 30×30 binary grid; the "Hard" designation applies to mazes with shortest path exceeding 110 steps, preventing greedy strategies. The dataset contains 1,000 training and 1,000 test samples. Each maze is encoded as a flattened binary sequence of length 900 (start=2, exit=3). Training mazes receive the 8 dihedral group transformations for augmentation, applied after the train/test split.

3.3 Dataset Justification

Both datasets appear in [Jolicoeur-Martineau, 2025; Dillon, 2026], enabling direct comparison without dataset confounds. They represent structurally different reasoning problems (fixed-size constraint satisfaction vs. variable-length pathfinding) in a data-scarce regime (1,000 training examples each), both fully synthetic and publicly available.

4 Methods

We implement and compare three model families on Sudoku-Extreme and Maze-Hard: TRM (fine-tuned from the HuggingFace checkpoint), four LLM baselines fine-tuned via LoRA, and two distilled student models. LLM baselines are deliberately limited to four decoder-only models spanning 124M–1B parameters to test whether failure scales with capacity. The TRM architectures and datasets are reused from [Jolicoeur-Martineau, 2025] via public HF mirrors; our engineering covers the LLM LoRA pipelines, the distillation pipeline and 2.4M student, CodeCarbon instrumentation across all families, the K-vote framework, and the `mask_non_path` bug fix found through independent re-implementation. All experiments ran across six identical machines (Intel Core i7-14700, NVIDIA RTX 5070 12 GB, 64 GB RAM, Windows 11) with CodeCarbon v3.2.6 [Courty et al., 2023] tracking environmental cost throughout.

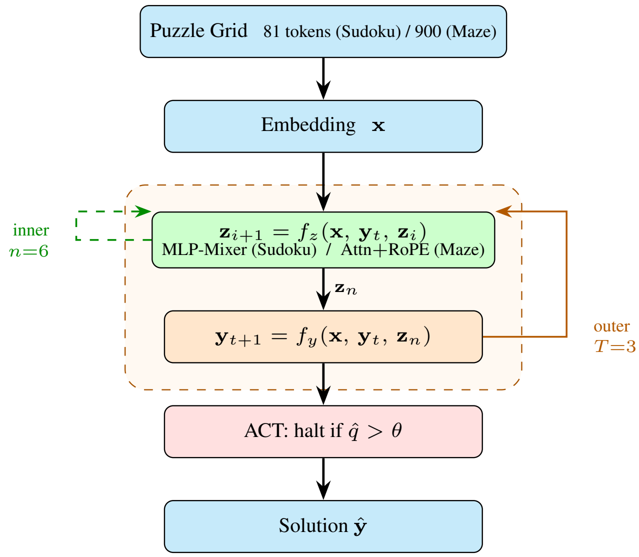

Figure 1: TRM inference loop. Input tokens are embedded into fixed features $\mathbf{x}$. The inner loop (green, n=6 steps) repeatedly refines reasoning features $\mathbf{z}$; the outer loop (orange, T=3 steps) updates the proposed solution $\mathbf{y}$ using the final $\mathbf{z}_n$. ACT halts early during training when confidence exceeds threshold θ; all T steps execute at test time.

4.1 Tiny Recursive Model (TRM)

Architecture

TRM follows Jolicoeur-Martineau [2025] with a single 2-layer network, hidden size $D = 512$, maintaining two latent states: proposed solution y and reasoning feature z. At each of $T = 3$ outer recursion steps, the network performs $n = 6$ inner latent recursion steps updating z given embedded input x, current y, and previous z; then y is updated. This gives 42 effective layers at only 5–7M parameters. We use **TRM-MLP** for Sudoku-Extreme (MLP-Mixer [Tolstikhin et al., 2021] over all 81 token positions, naturally capturing global constraint interactions) and **TRM-Att** for Maze-Hard (multi-head self-attention with RoPE for the 900-token context). Both use RMSNorm [Zhang and Sennrich, 2019], no bias, and SwiGLU activations [Shazeer, 2020]. ACT with $N_{\text{sup}} = 16$ supervision steps halts training via a BCE classification head on y, reducing average training steps to $\sim$2 on Sudoku while all 16 steps run at test time (Figure 1).

Training Protocol

TRM-MLP is fine-tuned from the published checkpoint of Jolicoeur-Martineau [2025][1] on Sudoku-Extreme using AdamW [Loshchilov and Hutter, 2017] (lr 1×10^{-4}, weight decay 1.0, batch 768), linear warmup over 2,000 iterations, up to 1,000 epochs with early stopping at the peak validation checkpoint (`best.pt`). The per-step objective is $\mathcal{L} = \text{CE}(\hat{y}, y_{\text{true}}) + \text{BCE}(\hat{q}, \mathbf{1}[\hat{y} = y_{\text{true}}])$ with the stablemax CE variant [Prieto et al., 2025] for numerical stability; an EMA (decay 0.999) is applied to all validation checkpoints. Our fine-tuned model achieves 74.54% $\pm$ 0.33% puzzle accuracy and 85.94% $\pm$ 0.16% cell accuracy (mean over 3 seeds); the same HuggingFace checkpoint evaluated directly on our test split achieves 84.74%, and the 10.20 pp gap is attributable to the 6$\times$ batch-size reduction (768 vs. 4,608 original) destabilising EMA convergence. For Maze-Hard, we evaluate the published TRM-Att checkpoint on our test split without further training, reporting 79.60% puzzle / 97.54% cell accuracy as the primary result.

4.2 LLM Baselines

We evaluate four decoder-only LLMs spanning 124M–1B parameters — GPT-2 Small (124M) [Radford et al., 2019], SmolLM2-360M (360M) [Allal et al., 2025], Qwen2.5-0.5B (500M) [Qwen Team, 2024], Llama-3.2-1B (1B) [Dubey et al., 2024] — to test whether LLM failure on structured reasoning is a capacity or an architectural issue. All four are fine-tuned with LoRA [Hu et al., 2022] (rank $r = 8$, $\alpha = 16$, applied to query and value projections), AdamW lr 5×10^{-5}, batch 16, cosine LR with 0.1 warmup ratio. GPT-2, SmolLM2, and Llama train for 30 epochs; Qwen for 100. Loss is computed only over solution tokens. Sudoku puzzles are serialised as 81-character digit strings; Maze as 900-character row-major sequences. Under the default `mask_non_path: true` evaluation, all four LLM and distilled student models (Qwen2.5-0.5B, GPT-2 Small, Distill-Qwen, Distill-GPT-2) score 100% on Maze-Hard, but this is a saturation artefact: the metric excludes wall cells ($\approx$80% of the grid), so any model that outputs a path-shaped sequence scores perfectly regardless of correctness. Re-evaluation with `mask_non_path: false` drops all to 0% puzzle; cell accuracy ranges from 12.50% (random floor) to 21.38% (GPT-2 Small).

4.3 Distilled LLMs

We train two distilled students from Qwen2.5-0.5B and GPT-2 Small respectively. Both share the same architecture: a 4-layer, 8-head, hidden-size-256 decoder ($\sim$2.4M parameters), initialised randomly. The KD objective is $\mathcal{L}_{\text{KD}} = \alpha \cdot \text{CE}(\text{logits}_S, y_{\text{true}}) + (1-\alpha) \cdot T^2 \cdot \text{KL}(\sigma_S \| \sigma_T)$ with $T = 4$, $\alpha = 0.5$, and T^2 correcting for temperature scaling [Hinton et al., 2015]. Students train for 30 epochs (AdamW, lr 5×10^{-5}) on the same 1,000-sample augmented set, teacher frozen. Instantiating both students identically isolates the effect of teacher quality on distillation gain.

4.4 Environmental Measurement and K-vote

CodeCarbon v3.2.6 [Courty et al., 2023] tracks GPU (NVML), CPU (Intel RAPL), and RAM power draw against the England UK grid intensity ($\approx$0.238 kgCO_2eq/kWh), logging training energy (kWh), CO_2 (kgCO_2eq), duration, and inference energy per puzzle (μkWh/puzzle). Following Dillon [2026], we also evaluate K-vote majority voting at $K \in \{1, 2, 4\}$ for TRM-MLP/Sudoku, TRM-Att/Maze, and all LLM baselines; results in Section 5.3.

[1] Public HF mirrors (uploaded by Sanjin2024): https://huggingface.co/Sanjin2024/TinyRecursiveModels-Sudoku-Extreme-mlp; https://huggingface.co/Sanjin2024/TinyRecursiveModel-Maze-Hard.

Model	Params	Puzzle %	Cell %
TRM-MLP (HF ckpt)	5M	**84.74**	**91.55**
TRM-MLP (ours, $\pm\sigma$)	5M	74.54±0.33	85.94±0.16
Distill-Qwen	2.4M	0.00	35.87
Distill-GPT-2	2.4M	0.00	18.72
Qwen2.5-0.5B	500M	0.00	19.07
Llama-3.2-1B	1B	0.00	19.74
SmolLM2-360M	360M	0.00	14.07
GPT-2 Small	124M	0.00	12.29

Table 1: Sudoku-Extreme accuracy results.

4.5 Ethical and Societal Considerations

While TRM's small size and low inference cost make it attractive for deployment, this property raises distinct ethical responsibilities. First, our evaluation is limited to two synthetic benchmarks; a model validated only on Sudoku-Extreme and Maze-Hard must not be deployed in safety-critical settings (e.g., drug interaction checking, fraud detection) without domain-specific validation and regulatory approval — benchmark accuracy does not constitute clinical or legal certification. Second, energy inequality shapes reproducibility: the published result of 87.4% required 8×H200 GPUs (approximately $20,000/month in cloud compute), a resource accessible only to well-funded institutions; our 74.54% replication on consumer hardware demonstrates that the research gap between resource-rich and resource-constrained groups is not merely academic. Third, although TRM's compact size is environmentally favourable at inference, the same compactness makes it easier to embed in always-on edge devices, which may aggregate energy consumption at scale. Finally, the Sudoku-Extreme and Maze-Hard datasets are fully synthetic with no personal data; however, any real-world adaptation (e.g., routing, scheduling) would require careful assessment of whose constraints the model is optimising and who bears the cost of errors.

5 Experiments

We report puzzle accuracy (exact-match) and cell accuracy (fraction of cells correct), with all energy figures from CodeCarbon CSV logs.

5.1 Main Accuracy Results

Sudoku-Extreme

Table 1 summarises puzzle-level and cell-level accuracy on the Sudoku-Extreme test set (423,000 puzzles) for all models.

The TRM-MLP HF checkpoint achieves 84.74% puzzle accuracy, substantially outperforming all LLM baselines (0% puzzle accuracy across the board). Our fine-tuned replication achieves 74.54% ± 0.33% puzzle and 85.94% ± 0.16% cell accuracy (mean over 3 seeds; Figure 2); the 10.20 pp gap from the HuggingFace anchor is attributable to the batch-size constraint discussed in Section 4. Llama-3.2-1B (1B) performs no better than GPT-2 Small (124M) at the puzzle level, confirming the failure is architectural, not a capacity issue. The distilled Qwen student (2.4M) reaches 35.87% cell accuracy, nearly double its teacher's 19.07%, suggesting soft-target distributions carry substantially more signal than hard labels.

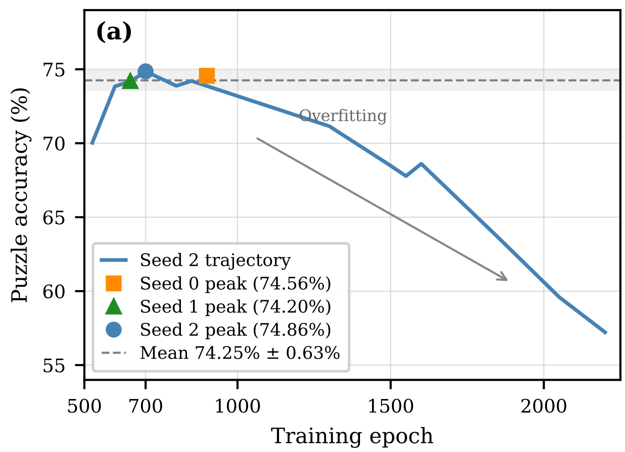

Figure 2: Validation puzzle accuracy vs. epoch for TRM-MLP fine-tuned from the HuggingFace checkpoint (seed 2 trajectory shown; peak markers for all three seeds). The model peaks at 700–900 epochs then overfits; mean 74.54% ± 0.33% across seeds.

Model	Puzzle %	Cell %	Note
TRM-Att (HF)	**79.60**	**97.54**	Full-grid
Qwen2.5-0.5B	0.00	12.52	Re-eval
GPT-2 Small	0.00	21.38	Re-eval
Distill-Qwen	0.00	12.50	Re-eval
Distill-GPT-2	0.00	12.50	Re-eval
TRM-Att (fine-tune)	≈0.00	—	Collapsed

Table 2: Maze-Hard accuracy under full-grid evaluation (`mask_non_path: false`). All four LLM/distill models scored 100% under the default mask (saturation artefact; see text). GPT-2 Small's 21.38% cell accuracy exceeds the ≈12.5% random baseline, indicating partial local structure learning; Distill-GPT-2 and Distill-Qwen are at the random floor.

Maze-Hard

Table 2 summarises results on the Maze-Hard test set (1,000 mazes) under full-grid evaluation (`mask_non_path: false`).

TRM-Att (HF, `mask_non_path: true`) achieves 79.60% puzzle / 97.54% cell accuracy; re-evaluated under the strict `mask_non_path: false` setting used for the LLMs below, TRM-Att still scores 78.90% / 99.30%, so the qualitative gap holds. Under that strict setting all four LLM/distill baselines collapse to 0% puzzle accuracy; cell accuracies are 12.52% (Qwen2.5-0.5B), 21.38% (GPT-2 Small), and 12.50% for both distilled students. GPT-2 Small's 21.38% is above the ≈12.5% random baseline (6-symbol vocabulary), indicating partial local structure learning from LoRA fine-tuning. Sudoku-Extreme results (full 81-cell scoring throughout) serve as the definitive cross-architecture comparison; future Maze-Hard evaluations should use full-grid scoring by default.

Model	Energy (kWh)	CO_2 (kg)	Duration
Qwen2.5-0.5B	0.8960	0.2129	∼6.8 hr
Llama-3.2-1B	0.5813	0.1381	∼2.6 hr
SmolLM2-360M	0.3043	0.0723	∼1.6 hr
GPT-2 Small	0.2570	0.0611	∼55 min
Distill-Qwen	0.0092	0.0022	∼3 min

Table 3: Training energy and emissions—Sudoku-Extreme.

Configuration	Eff. depth	Puzzle %	Params
TRM (T=3, n=6)	42	**87.4**	5M
no EMA	42	79.9	5M
T=2, n=2 (reduced)	12	73.7	5M
HRM (baseline)	24	55.0	27M

Table 4: Ablation of TRM-MLP on Sudoku-Extreme (replicated from Jolicoeur-Martineau [2025], Table 1).

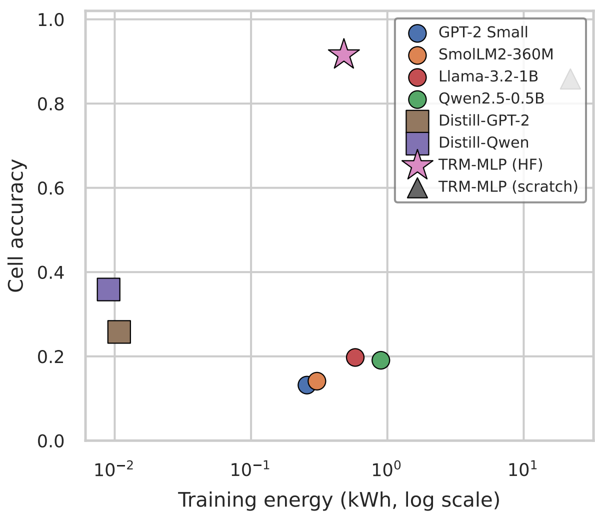

Figure 3: Training energy (kWh, log scale) vs. cell accuracy for all Sudoku-Extreme models. Distill-Qwen (square) achieves the best cell accuracy at near-zero training cost. TRM-MLP (HF, star) requires no fine-tuning on our hardware.

5.2 Environmental Cost

Table 3 reports training energy and CO_2 for all models on Sudoku-Extreme, from CodeCarbon CSV logs on identical hardware in England, UK.

The distilled Qwen student is trained at 0.0092 kWh (approximately 30× less training energy than an epoch-matched 30-epoch Qwen LoRA run at 0.272 kWh) while achieving nearly double the teacher's cell accuracy (35.87% vs. 19.07%), as illustrated in Figure 3.

The student-beats-teacher result has a mechanistic explanation: Sudoku outputs have combinatorial structure, so the Qwen teacher's soft targets expose near-miss cell assignments that hard labels discard — a lower-entropy signal consistent with Hinton et al. [2015]. The student learns which assignments are *close to* correct, providing a richer inductive bias than one-hot supervision [2019]. The Distill-GPT-2 student (weaker teacher: 13.18% cell accuracy) gains only 12.62 pp vs. Distill-Qwen's 16.80 pp, confirming that a stronger teacher produces richer soft targets.

The training-energy / accuracy relationship across the LLM family is non-monotonic: Llama-3.2-1B (0.5813 kWh) matches GPT-2 (0.2570 kWh) at the puzzle level, and Qwen2.5-0.5B, the most expensive model (0.8960 kWh, 6.8 hr), is nearly halved on cell accuracy by its 3-minute distilled student. Per-puzzle inference cost is highest for TRM-MLP (1.02 μkWh, 19.9 ms) due to 16 recursive steps; Distill-GPT-2 is below resolution (<0.01 μkWh, 0.13 ms). TRM-Att costs 7.20 μkWh per maze puzzle at 122.8 ms.

5.3 K-vote Test-Time Compute

K-vote majority voting at $K \in \{1, 2, 4\}$ produced no monotonic accuracy gain for TRM models. TRM-MLP fluctuated (54.37% → 51.62% → 52.65%) and TRM-Att was flat (79.60% / 79.40% / 79.60%), while inference energy scaled linearly with K (TRM-MLP: 1.02 → 4.40 μkWh). LLM baselines scored 0% puzzle regardless of K; an extended sweep to $K = 16$ for Qwen2.5-0.5B yielded marginal gains (15.76% → 17.42%), and a $K \in \{1, 2, 4, 8\}$ sweep for Distill-Qwen (correct machine-1 checkpoint, 35.87% cell at argmax; $T = 0.7$) showed a 4.83 pp gain from $K = 1$ to $K = 8$ (21.36% → 26.19%) — the largest K-vote improvement observed, yet still below the argmax baseline and without crossing the 0% puzzle threshold. The TRM-MLP $K = 1$ value (54.37%) uses the epoch-150 snapshot of an iso-time HF-init fine-tune (peaked 84.84% at epoch 10, declined due to overfitting); the 84.74% in Table 1 is the published HF checkpoint evaluated separately. The Distill-Qwen gain is consistent with Dillon's [2026] finding that large gains require latent diversity internalised during training — soft-distilled student models can still benefit modestly from test-time aggregation, but the effect is small relative to the argmax-to-stochastic gap.

5.4 Ablation: Recursion Depth and Iteration Limits

Table 4 replicates key ablations from Jolicoeur-Martineau [2025] (Table 1) to contextualise our training results.

Our fine-tuned replication (74.54% ± 0.33%) falls between the "no EMA" (79.9%) and "T=2, n=2" (73.7%) ablations, consistent with the batch-size constraint limiting convergence rather than misconfiguration; the 12.86 pp gap to the published result (87.4%) is therefore attributable primarily to batch size. The ablations confirm three findings that bear on our results: EMA is critical (87.4 → 79.9% if removed), recursion depth matters (halving depth drops to 73.7%), and MLP-Mixer outperforms self-attention by ≈12.7 pp on Sudoku.

6 Conclusion

This project reproduces and extends the Tiny Recursive Model (TRM) of Jolicoeur-Martineau [2025] on Sudoku-Extreme and Maze-Hard under consumer-GPU constraints. TRM-MLP (HF) achieves 84.74% on Sudoku; TRM-Att

79.60% on Maze-Hard; GPT-2, Qwen2.5-0.5B, and Llama-3.2-1B all score 0% puzzle-level regardless of size; our fine-tuned replication reaches 74.54% $\pm$ 0.33%, with the gap attributable to the batch-size constraint. Knowledge distillation produces a surprising secondary result: the distilled Qwen student (2.4M) achieves 35.87% cell accuracy, nearly double its teacher's 19.07%, suggesting soft target distributions carry more signal than hard labels.

Environmentally, the distilled student requires 0.0092 kWh ($\approx$30$\times$ less than an epoch-matched Qwen LoRA run) while achieving nearly double the cell accuracy. K-vote yields no gain for TRM or pure LLM baselines; Distill-Qwen gains 4.83 pp at $K = 8$ (21.36% $\rightarrow$ 26.19%), still below its 35.87% argmax baseline, confirming that large K-vote gains require latent diversity internalised during training.

TRM's advantages are most pronounced in settings requiring exact logical correctness — drug interaction checking, real-time fraud detection, chip design verification — exactly where LLMs fail and embedded-hardware energy budgets rule out large models. Rauba et al. [2026] show that adapting TRM to autoregressive settings yields no reliable gains, reinforcing that its advantage is specific to constrained, non-autoregressive problems.

The primary limitation is the batch-size constraint: the published TRM was trained at 4,608 samples/batch on 8$\times$H200, while our hardware supports a maximum of 768, a 6$\times$ gap that is the dominant cause of the 10.20 pp drop from the HuggingFace anchor. LLM fine-tuning runs used a single seed each; variance estimates are unavailable for these baselines. Future work should investigate gradient accumulation and larger datasets to improve consumer-GPU fine-tuning, and all Maze-Hard evaluations should use full-grid scoring by default.

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